

ECM3446

UNIVERSITY OF EXETER

**FACULTY OF ENVIRONMENT, SCIENCE
AND ECONOMY**

COMPUTER SCIENCE

Examination, May 2024

High Performance Computing

Module Leader: Man Luo

Duration: TWO HOURS

No Word Count Specified

Answer ALL questions.

The marks for this module are calculated from 60% of the percentage mark for this paper plus 40% of the percentage mark for associated coursework.

Candidates are permitted to use approved portable electronic calculators in this examination.

This is a CLOSED BOOK examination.

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Question 1

- (a) You are given the task of helping a researcher speed up their program and you suggest parallelising the program. The researcher says: “Parallel programming is too difficult. I’ve heard that Moores Law is still going. Why can’t I get a much faster processor to make my serial program run quicker?”

How would you explain to them that they have mis-understood Moores Law and why they can’t get a faster single processor core?

(8 marks)

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(b) The Top500 list ranks the fastest supercomputers in the world according to the performance on the LINPACK benchmark.

(i) What is the LINPACK benchmark?

(2 marks)

(ii) Explain the reasons behind the common observation that real-world applications often exhibit lower performance levels compared to the LINPACK benchmark.

(6 marks)

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- (c) The 3rd fastest computer on the June 2012 Top500 list was called the Mira. The following table shows information about the Mira taken from the Top500 website.

Manufacturer:	IBM
Cores:	786,432
Processor:	Power BQC 16C 1.6GHz
Interconnect:	Custom Interconnect
Linpack Performance (Rmax):	8,586.61 TFlop/s
Theoretical Peak (Rpeak)	10,066.33 TFlop/s
HPCG performance:	167.049 TFlop/s
Power consumption:	3,945.00 kW
Operating system:	Linux

- (i) What is the LINPACK efficiency of the Mira? Show your working.
(2 marks)
- (ii) What is the theoretical peak performance of a single processor core?
(4 marks)
- (iii) Based on the information shown above would you class the Mira as a commodity cluster or an MPP (Massively Parallel Processor) and why?
(2 marks)

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(d) Matrix operations are often used in high performance computing to solve systems of linear equations.

(i) What is the difference between a sparse matrix and a dense matrix?

(2 marks)

(ii) Describe the compressed sparse row format for storing sparse matrices.

(4 marks)

(iii) How does the number of bytes of data transferred from memory scale with N for dense matrix multiplication and why?

(2 marks)

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- (e) MPI is a widely used parallel programming technology which can be used to write parallel programs to run on a wide range of architectures. OpenMP is another parallel programming technology which can be used instead of, or in combination with, MPI. Describe four situations where you would choose to use OpenMP instead of, or in combination with, MPI.

(8 marks)

(Total 40 marks)

Question 2

- (a) The following C program makes a call to the MPI (Message Passing Interface) library but the call has been removed and replaced with the “Missing line” comment.

```
#include <stdio.h>
#include <mpi.h>

int main(int argc, char *argv[]){

    int a[3], b[6];
    int myRank, i;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &myRank);

    for (i=0; i<3; i++){
        a[i] = i + myRank*3;
    }

    for (i=0; i<6; i++){
        b[i] = 0;
    }

    printf("Rank: %d a before: %d %d %d\n",
           myRank, a[0], a[1], a[2]);
    printf("Rank: %d b before: %d %d %d %d %d %d\n",
           myRank, b[0], b[1], b[2], b[3], b[4], b[5]);

    /***** Missing line *****/

    printf("Rank: %d a after:  %d %d %d\n",
           myRank, a[0], a[1], a[2]);
    printf("Rank: %d b after:  %d %d %d %d %d %d\n",
           myRank, b[0], b[1], b[2], b[3], b[4], b[5]);

    MPI_Finalize();

    return 0;
}
```

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When the program is run using 2 MPI processes it produces the following output

```
Rank: 0 a before: 0 1 2
Rank: 0 b before: 0 0 0 0 0 0
Rank: 1 a before: 3 4 5
Rank: 1 b before: 0 0 0 0 0 0
Rank: 1 a after:  3 4 5
Rank: 1 b after:  0 0 0 0 0 0
Rank: 0 a after:  0 1 2
Rank: 0 b after:  0 1 2 3 4 5
```

Write out the missing MPI call making sure to include all the arguments.

(10 marks)

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(b) You are providing programming support to a researcher who is using a C program to perform calculations. They are running the program overnight on a laptop computer and it takes 8 hours to complete. They need to get results from their program more quickly, ideally within an hour. When you look at their program you see that it is a serial program with no parallelisation, and most of the work is in `for` loops. You know that there is a server with a single 12-core processor which could be used to run the program. You decide to parallelise the program so that it can run on the server.

- (i) What is the minimum parallel efficiency required on this server to reduce the 8 hour run time to 1 hour? Show your working.

(4 marks)

- (ii) You need to deliver a working parallel program as quickly as possible. Would you choose OpenMP or MPI for parallelising the program? Explain your choice.

(6 marks)

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- (c) Show how the following sparse matrix would be represented in compressed sparse row format with rows and columns numbered starting from one.

(10 marks)

$$\begin{pmatrix} 5 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 7 \end{pmatrix}$$

(Total 30 marks)

Question 3

- (a) Consider a two-dimensional partial differential equation (PDE) solver which solves a PDE with one unknown variable u . To solve the PDE it is necessary to calculate values for the second derivative

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

This expression is evaluated using the finite difference stencil

$$\text{del2u} = (u(i+1, j) - 2*u(i, j) + u(i-1, j)) / (dx*dx) + (u(i, j+1) - 2*u(i, j) + u(i, j-1)) / (dy*dy)$$

The computation domain for the whole calculation comprises 8192×8192 grid points. In addition to these grid points each sub-domain stores halo values which are copies of the data on neighbouring processors.

- (i) The calculation is domain decomposed in an 16×16 configuration such that there are 256 square sub-domains of equal size. How many values need to be exchanged when one halo is sent to one neighbouring process?

(2 marks)

- (ii) How many bytes need to be sent if the values are represented using IEEE754 double precision?

(2 marks)

- (iii) What is the transmission time for sending a single halo over a network with a latency of $1\mu\text{s}$ (10^{-6} s) and a bandwidth of 15 GB/s (15 gigabytes per second)? Show your working.

(3 marks)

- (iv) The domain decomposition is modified so that the domain is decomposed in a 32×32 configuration such that there are 1024 square sub-domains of equal size. What is the transmission time for one halo with the modified domain decomposition?

(3 marks)

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(b) The “Memory Intensive Service” is an HPC facility hosted by the University of Durham at the Institute for Computational Cosmology (ICC). The system comprises 452 compute nodes, each with 512 GB of RAM, and offering a total of 12 656 cores. All the compute nodes have the same number of processor cores.

(i) What is the largest number of processor cores available to an OpenMP program running on this system and why?

(3 marks)

(ii) What is the largest number of processor cores available to an MPI program running on this system and why?

(3 marks)

(iii) Unified Parallel C can be used to write parallel programs as an alternative to OpenMP and MPI. According to Wikipedia:

Unified Parallel C (UPC) is an extension of the C programming language designed for high-performance computing on large-scale parallel machines, including those with a common global address space (SMP and NUMA) and those with distributed memory (e.g. clusters). The programmer is presented with a single shared, partitioned address space, where variables may be directly read and written by any processor, but each variable is physically associated with a single processor. UPC uses a single program, multiple data (SPMD) model of computation in which the amount of parallelism is fixed at program startup time, typically with a single thread of execution per processor.

Based on this information what is the largest number of processor cores available to a Unified Parallel C program running on this system and why?

(4 marks)

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(c) Performing large scale calculations requires the use of parallel programs which can scale effectively to use many processors. Understanding the efficiency and scaling properties of a parallel program is important in order to make effective use of HPC systems.

- (i) Write down an expression for Amdahl's law and say which type of scaling (strong or weak) it describes. You should clearly define the symbols used in the expression.

(3 marks)

- (ii) According to Amdahl's law, what is the maximum possible speedup of a parallel program with a parallel fraction p ?

(2 marks)

- (iii) Write down an expression for Gustafson's law and say which type of scaling (strong or weak) it describes. You should clearly define the symbols used in the expression.

(3 marks)

- (iv) How does run time vary with number of processors for an application which exhibits ideal weak scaling?

(2 marks)

(Total 30 marks)